ROS2 Installation Instructions

Dear Students,

In this lab, we are going to use a middleware called ROS 2. This middleware supports a variety of OS, but we can provide assistance only for Ubuntu 24.04, that's our choice for this lab. If you feel confident enough, you could try to install ROS 2 in your current OS, but we cannot guarantee that everything will work as intended.
There are various possibilities to install the linux version:

- Bare metal on your laptop (in dual booting with your other os) is the best choice for performance, but you must pay attention to the installation procedure if you do not want to lose your data.
- A full Virtual Machine, as with VirtualBox or VMWare, is the best for compatibility, but the performance could not be good, especially for the robotic simulation if your pc is not powerful enough.
- A docker/podman container if you are nerd enough.
- In case all these choices are not suitable for your pc, the university provides a compatible remote machine for this course (The performance may not be optimal if you are all connected at the same time).

Your tutors

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Hardware Suggestion:

- Powerful enough multi-core CPU (at least Intel i5 6th generation or equivalent)
- A discrete GPU is not mandatory, but recommended for VM users
- at least 8 GB of RAM

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Installation:

Ubuntu 24.04:
https://ubuntu.com/download/desktop

then,
ROS2 Jazzy Jalisco:
https://docs.ros.org/en/jazzy/Installation.html
choose Binary Package -> Ubuntu Linux -> deb packages
Follow the sequence of instructions and commands until 'try some examples'.

The following additional packages will be used in the course so please install them:

```
sudo apt-get -y install ros-jazzy-ament-cmake \
ros-jazzy-ur ros-jazzy-ur-simulation-gz ros-jazzy-navigation2
ros-jazzy-nav2-bringup   \
ros-jazzy-nav2-minimal-tb* ros-jazzy-turtlebot3-gazebo
ros-jazzy-rviz-visual-tools \
```

```
ros-jazzy-gz-ros2-control ros-jazzy-ros-gz-sim
ros-jazzy-joint-state-broadcaster \
ros-jazzy-joint-trajectory-controller ros-jazzy-pcl-ros
ros-jazzy-realsense2-description  \
ros-jazzy-robotiq-controllers ros-jazzy-robotiq-description
ros-jazzy-ros-gz \
ros-jazzy-apriltag ros-jazzy-apriltag-ros ros-jazzy-apriltag-detector \
ros-jazzy-image-proc ros-jazzy-depth-image-proc ros-jazzy-nav2-route
```

Discovery settings

If you see other people's topics, you should modify the discovery setting according to

https://docs.ros.org/en/jazzy/Tutorials/Advanced/Improved-Dynamic-Discovery.html#improved dynamicdiscovery

at the end of your .bashrc add

```
export ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST
```

Extra tip - 1

A super-easy wrapper for a container is distrobox. With this, you can create a container with:

```
distrobox create -n uburos -i ubuntu:24.04
```

And then, all the time you need to enter the container, you only need to use:

```
distrobox enter uburos
```

Distrobox automatically shares your home directory and the network pipeline of your pc

Extra tip - 2

You may need an updated driver for your graphics card if you are using podman/docker (check this online before using this):

```
sudo add-apt-repository ppa:kisak/kisak-mesa
sudo apt update
sudo apt upgrade
```

Windows Subsystem for Linux (Windows Users)

For Windows users, an alternative to installing a VM is **WSL (Windows Subsystem for Linux)**, which allows you to run a full Linux environment directly on Windows **without needing a separate virtual machine or dual boot**. To set it up, open **PowerShell as Administrator** and run:

```
wsl --install -d Ubuntu-24.04
```

This will install WSL and Ubuntu 24.04 automatically. Once the installation completes, **restart your computer**, then launch **Ubuntu** from the Start menu or by opening a terminal and running the command: `ubuntu2404`

You'll be prompted to create a **Linux username and password**. After setup, update your system with:

```
sudo apt update && sudo apt upgrade
```

You now have a fully functional Ubuntu environment ready for development directly within Windows.

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UTM virtual machine - Virtualization of Linux under Apple ARM devices

Change the virtualization configuration from

```
QEMU 10.0 ARM Virtual Machine (virt-10.0)
```

to

```
Linux
```